Flight Dynamics of the Sud Aviation SA 316B Alouette III in JSBSim

An articulated-rotor blade-element model with a single-shaft turboshaft, built from public data and flown through a scripted test programme

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Abstract

A flight-dynamics model of the Alouette III for the FlightGear simulator has been written in JSBSim, replacing the model built on JSBSim's generic rotor that the aircraft flew before. Its fully articulated rotor is simulated blade by blade: three rigid blades on coincident flap and lag hinges (the lag frequency 0.183 per revolution), with a viscous lag damper, eight segments each on tables of its NACA 63A011.5 section built from Boeing Vertol's C-81 equations for the 0012 and like-for-like wind tunnel data, under a Peters–He finite-state wake with Johnson's vortex ring model, ground effect and a ring-vortex wake on the tail surfaces. The Turbomeca Artouste III B is modelled as what it is, a single-shaft engine whose spool is geared to the rotor through a centrifugal clutch and a freewheel, with its isochronous governor, its starter and its inertia. Since the Alouette's rotor turns clockwise, the aerodynamics are worked in the aircraft's mirror image. The model, 581 aerodynamic functions and 256 control-system components, is generated from transcribed data and runs in plain JSBSim at 480 Hz. No flight-test time histories of the Alouette III are public, so the scripted programme checks what can be checked: the published performance, the published start, the rotor's physics against theory, and a test flight flown by an autopilot from lift-off to touchdown. The model hovers at 2200 kg on 364 kW and reaches the 405 kW continuous rating at 208 km/h against the published maximum of 210 km/h. The published hover ceilings imply that only 342–378 kW reach the rotor, and at that power the model climbs at 4.5–5.9 m/s against the published 4.3–5.5 m/s: the published performance is consistent with itself and with the model only at about nine-tenths of the rated power, a discrepancy left open rather than tuned away. The start runs as published (idle at 16330 rpm against 15 000–18 000, the rotor taken up at 20823 rpm of the engine against 19 500–21 000), and in vertical descent the heave mode is unstable from 869 to 2819 ft/min, the vortex ring state.

1 Introduction

The Alouette III distributed for FlightGear (by G. Robin and others, with later systems work) is a detailed aircraft: cockpit, systems, a hoist, a co-pilot who hovers for it. Its flight model used JSBSim's built-in rotor, which computes a rotor's thrust and moments from a few parameters in closed form; it flew, but not like a helicopter of the Alouette's kind. The Alouette III is a light utility helicopter of 1959: a fully articulated three-bladed rotor turning clockwise seen from above (the French practice), a three-bladed tail rotor on the right of an open lattice boom, and a single-shaft Turbomeca Artouste turboshaft whose compressor and turbine turn at constant speed and drive the rotor through a centrifugal clutch.

Unlike the Bo 105 and UH-60A, whose models this one follows and which had flight tests and full analysis decks to be checked against, the Alouette III has no published flight-test time histories, identified derivatives or blade structural data. What is public is its type certificate, the engine's, performance figures in several sources (which disagree), a few technical papers, a fuselage wind tunnel test, tyre data, and descriptions of the start of its sister, the Lama (Table 1). This report describes a model built on that, after the Bo 105 and UH-60 models, and a test programme that checks it where the data allow; where they do not, it says so. All result tables, figures and the numbers quoted from them are written by the evaluation scripts.

2 Reference data

A few points of data reduction matter for the comparisons. *(i)* The sources' performance figures disagree. At 2200 kg This Day in Aviation gives hover ceilings of 1650 m in ground effect and 100 m out of it, Aérostories 2850 and 1100 m; the cruise speed is 180, 185 or 195 km/h. All are read as sea level, standard day. *(ii)* The blade's mass and inertia are not published. The Alouette II's metal blade, of the same construction, weighs 28.9 kg for 4250 mm of span and 280 mm of chord [8]; scaled to the Alouette III's span and chord it gives 38.7 kg, a Lock number of 5.1. *(iii)* The blade's twist, 6°30′, and its section, NACA 63A of 11.5 % thickness, are Hindustan Aeronautics' description of the type [13], which names no camber: the section is taken as the symmetrical 63A011.5. The tail rotor's sections are NACA 0015 to 0018, taken as 0018 at the root and 0015 at the tip. No full tables of these sections, over the whole circle and up to the advancing tip's Mach number, are published; they are built from C-81's 0012 (Section 3.5). *(iv)* The hinge offset, the blades' aerodynamic span and the tail surfaces come from the FlightGear 3-D model, whose blade animation pivots on the hinges. *(v)* The control ranges are not published: the type certificate refers to the maintenance manual for the rigging. They are estimates wide enough to trim from the hover to beyond V_{NE}.

3 The model

3.1 Implementation

A Python script writes the model from the transcribed data (Table 2). JSBSim evaluates aerodynamic functions on demand, so the function graph must be free of cycles; every quantity that depends on its own past (the blades' flap and lag angles and rates, the rotor's azimuth and speed, the wake states, the tail rotor's flapping and inflow, the engine's spool, fuel and temperature) lives in control-system channels, integrated explicitly. The model runs at 480 Hz, 4.4° of the ro-

Table 1: Sources and what they contribute.

Source	Content used
EASA TCDS R.123 [1]	dimensions, rotor speeds and limits, masses, CG range and datum, fuel, transmission limits, V_{NE}
EASA TCDS E.091 [2]	Artouste III B: stages, speeds, ratings, T4 limits, reduction gear, mass
This Day in Aviation [3]	rotor senses and speeds, tail rotor; speeds, hover ceilings, climb at 2200 and 1750 kg
Aérostories [4]	specification table of the Artouste version: speeds, climb, hover and service ceilings, range
heli-archive.ch [5]	rotor and engine description; the gearbox-limited power held to 4000–4500 m; fuel flow
2nd ERF [6]	the rotor head's mass
3rd ERF [7]	the Artouste's isochronous governor and automatic start
14th ERF [8]	the Alouette II's metal blade: mass, chord, section, twist
20th ERF [9]	the Alouette III fuselage in the wind tunnel: lift and drag against angle of attack
Michelin, Goodyear [10]	the 355×150-4 tyres: loads, deflection, sizes
ROTORwrench [11]	the Lama's Artouste IIIB1: idle, the starter's cut-out
Lama add-on manual [12]	a flight simulator's, after the flight manual: the engine speed at which the rotor turns, the rotor brake
HAL description [13]	a licence-built SA 316B: blade chord, section and twist, lag dampers, tail rotor sections, fuel burn
RST 1/2020 [14]	the blade's chord along the span, after the pilot's handbook
TR 75-50 [16]	C-81's NACA 0012 equations, the sections' base
NACA 903, 824 [17, 18]	the 63A010 and 63_1-012 in the Langley tunnel; critical Mach numbers
NASA TM 4074 [19]	the 0012 in the same tunnel, smooth and rough
NACA 460 [20]	the 0012, 0015 and 0018: slopes, maximum lift, minimum drag
Sound recordings	the tail rotor's gearing, from its blade-passing frequency; the start's pace, idle, the clutch's slip
AOPA Pilot, 1972 [15]	a flight test of an SE 3160: autorotative descent, idle, the clutch
FlightGear 3-D model	hinge offset, blade span and chord, tail surfaces, wheels, lights
Johnson [21]	the vortex ring state model
NASA TN D-234 [22]	ground effect in forward flight

tor's azimuth and 24.2° of the tail rotor's per step; the airframe's loads, the engine and the outputs at 240 Hz, the original aircraft's systems (electrics, lights, gauges, its autopilot) at 120 Hz. The report's programme run at 960 Hz gives the same numbers, to within 1 % or their last digit, but for the start's clutch, which engages 0.7 s sooner at 480, and the last trim at 1750 kg, 5 kt further on the stall's edge; 400 Hz still holds them, while at 360 the clutch's run-up changes and at 320 the stopped rotor's friction no longer holds it. The rotor turns clockwise seen from above and the tail rotor sits on the right, pushing left: the mirror image of the UH-60's and Bo 105's layouts, whose rotor equations the generator shares.

Table 2: The generated model: JSBSim system files.

File	Content	Items
`al3-aero.xml`	the aerodynamic functions: blade segments, wake, tail rotor, the airframe; the six force and moment axes	581
`al3-rotor.xml`	blade flap and lag, rotor azimuth and speed, Peters–He wake states, the vortex ring, tail rotor flapping and inflow	82
`al3-airframe.xml`	fuselage, hub, gear and stabiliser loads, the ring-vortex wake on them, at 240 Hz	77
`al3-drive.xml`	engine, fuel control and governor, starter, clutch, freewheel, rotor brake, fuel	42
`al3-controls.xml`	pilot controls, the mirror image, rigging, servos	28
`al3-outputs.xml`	properties for the 3-D model and the instruments	27

Table 3: Main data (TC type certificate, TD This Day in Aviation, TE engine certificate, E88 14th ERF, AC a blade, VP Vertipedia, HL Hindustan Aeronautics' description, SR sound recordings' blade-passing frequencies, 3D the FlightGear 3-D model).

Item	Value	Source
Main rotor		
radius, blades, chord	5.51 m, 3, 0.350 m	TC, HL
speed, tip speed, solidity	353.2 rpm, 204 m/s, 0.0607	TD
twist, root cut-out, shaft tilt	-6.5°, 0.96 m, 2.5° fwd	HL, 3D
section	NACA 63A011.5	HL
flap/lag hinge offset	0.119 m (2.2 % R)	3D
blade mass, Lock number	38.7 kg, 5.1	E88 (est.)
flap, lag frequency	1.017, 0.183 /rev	derived
lag damper	25 % of critical	est.
Tail rotor		
radius, blades, chord	0.956 m, 3, 0.17 m	TD, AC
speed, sections	1938 rpm; NACA 0018–0015	SR, HL
Engine		
Artouste III B: speed, ratings	33500 rpm; 420, 405 kW	TE
reduction to the rotor	94.85:1	TE, TD
Mass		
empty, maximum	1143, 2200 kg	VP, TC

The aerodynamics are therefore worked in the aircraft's mirror image across its plane of symmetry (the lateral velocity, the roll and yaw rates and their accelerations, the lateral acceleration and the bank angle negated), and the side force, rolling and yawing moments negated back.

3.2 Main rotor

The three blades are rigid, on coincident flap and lag hinges 1.017 and 0.183 per revolution in flap and lag (Table 3), with the UH-60 model's flap and lag equations: the aerodynamic moments, the blade's weight and the hub's specific force, the gyroscopic and Coriolis terms, the centrifugal stiffness through the hinge offset, and the airframe's angular velocity and acceleration. A viscous lag damper gives 25 % of critical damping at the rotor's speed (an estimate); flap, droop and

lag stops limit the blades. Each blade, twisted 6.5° [13], has eight segments of equal annulus area outboard of its root, on the tables of its NACA 63A011.5 section (Section 3.5) at the segment's angle of attack and Mach number, with a drag increment of 0.0012 for a production blade and a 3 % tip loss. The hub's forces and moments come from the blades' loads at the hinges, their inertial reactions included, at the hub's position in the airframe. The rotor's speed is solved together with the three lag accelerations, as in the Bo 105 model: the blades' lag and the hub exchange angular momentum through the hinges.

3.3 Inflow and wake

The Peters–He finite-state wake has five states (four cosine harmonics, one sine), driven by the blades' segment lifts, with a time constant that follows the mass flow. In descent the mass flow parameter is Johnson's [21], as in the MV-22B model: through the vortex ring state the induced velocity rises above momentum theory's and the rotor loses thrust at a fixed collective, the heave mode going unstable. Ground effect scales the induced velocity with the hub's height as the Bo 105 and UH-60 models do, with Heyson's ratio into forward flight [22]; its strength is theirs, not fitted. A ring-vortex model of the main rotor's wake, precomputed against skew angle and the stations' positions, gives the downwash at the tail rotor, the stabiliser, the end plates, the fuselage and the boom.

3.4 Tail rotor

The three-bladed tail rotor is a blade-element rotor with six segments per blade on NACA 0018 (root) to 0015 (tip) tables, each blade on its own flapping hinge, with the 5° toe of its hub (3-D model) and a Pitt–Peters inflow. It is geared 5.487:1 to the main rotor, the ratio of the two rotors' blade-passing frequencies in seven recordings of governed Alouette III and Lama rotors, so that it turns at 1938 rpm. The published 2001 rpm [3, 13] is the tail rotor's speed at a main rotor's of about 365 rpm: Hindustan Aeronautics pairs it with 363 rpm [13], a ratio of 5.51, while the 353.2 rpm it is usually quoted with would make 5.67.

3.5 Blade sections

The sections' tables start from C-81's 0012 equations [16] (lift slope and maximum lift against Mach number, the drag polynomial and its rise, the flat plate past the stall, reversed flow) and take the differences that like-for-like tests give. In the Langley low-turbulence pressure tunnel the 0012 [19], the 63A010 [17] and the 63_1-012 [18] were measured smooth and with standard roughness (6A and 6-series sections of one thickness have the same maximum lift and drag [17]). The C-81 0012 lies between the smooth and the rough 0012: its maximum lift at Mach 0.15, 1.36, is two thirds of the way from the rough section's to the smooth one's at 3 million Reynolds number, the retreating blade's when it stalls, and its zero-lift drag with the production increment, 0.0092, nearly the rough section's at 6 million, the hovering and advancing blade's. The 63A011.5, interpolated in thickness, is placed the same way between its own smooth and rough values; the 0012 gains maximum lift with Reynolds number and the 63A hardly any. Its tables thus take 0.91 of the 0012's maximum lift up to Mach 0.3, going to the 0012's at Mach 0.6, where both sections' stall is the shock's and their critical Mach numbers at high lift are about the same [18] (an estimate: no high-speed data of a 63A section were found); add 0.0026 of drag at a lift of 0.6, growing with the angle of attack squared; and move the drag divergence Mach number up by 0.028, the difference of the two sections' critical Mach numbers at zero lift [18], fading out by 3.7°, where their curves meet. The lift-curve slopes are the same, 0.107 per degree. The tail rotor's 0015 and 0018 take their slopes, maximum lifts and minimum drags relative to the 0012 from the variable-density tunnel's series [20] (drag 12 and 30 % higher, maximum lift the same and 3 % lower), and their drag divergence 0.040 and 0.075 earlier, the 63-series' change of critical Mach number with thickness [18].

3.6 Airframe

The fuselage's lift and drag against angle of attack are the 20th ERF's wind tunnel data for the Alouette III [9], on the rotor disc's 95 m^2; side force and yawing moment by the cross-flow principle, with the body's Munk moments. The parts the tests leave out (the open tail boom, the gear, the rotor head and mast, and the rest) are cross-flow and flat-plate estimates; the last, 4 ft^2, is a round figure, not fitted. The stabiliser (0.8 m^2 between its end plates) and the end plates are lifting surfaces in the rotor's wake.

3.7 Flight controls

Collective, cyclic and pedals drive the swashplate and the tail rotor's pitch through hydraulic servos (the UH-60's response, for want of the Alouette's), with no augmentation. FlightGear's trims shift the controls' centres. The original aircraft's autopilot (heading and altitude holds) and its co-pilot for the hoist add to the pilot's controls, as they did before.

3.8 Engine, drive and mass

The Artouste III B is a single-shaft turboshaft: its compressor and turbine turn at 33 500 rpm, 94.85 times the rotor's speed, at whatever power [2, 7]. Its hydromechanical governor is modelled as an isochronous PI controller on the spool's speed, its fuel limited by the turbine temperature (T4, 550 °C at the thermal maximum). The spool's torque comes from the fuel beyond what it needs to turn unloaded; the thermal maximum, 640 kW at sea level, lapses with altitude so that the gearbox's 420 kW is held to 4250 m [5]. The start follows the published descriptions [11, 12] and, in its pace, two recorded starts: the starter cranks the spool, its torque falling to nothing at 17 % as an electric starter's, the spool at 10 % in 4 s; the start system's fixed fuel flow until idle; light-off at 16 %, about 14 s after the starter, which goes off line at 14 000 rpm; idle at 16 330 rpm (a cockpit recording's; about 15 000 [15], and 17 000–18 000 for the Lama's engine [11]) about 30 s after light-off, with the fuel flow lever at 35 %. Advancing the lever, the centrifugal clutch slips from 19 000 rpm [15] with its shoes' sliding friction, fitted to the recordings' take-ups (the rotor from rest at flat pitch, the spool held): about 1150 ft lb at the rotor at 62 % of the spool's speed and 1490 at 80 %, so that the rotor locks to a spool at 80 % in 44 s (45 recorded) and at 61 % in 62 s (64 at 63 %), and above as a centrifugal clutch's, with the speed

squared. Locked, its capacity, full by 24 000 rpm, carries any flight torque. A freewheel lets the rotor overrun the engine. The spool's inertia, 0.15 kg m^2, is 1350 kg m^2 at the rotor's speed, as much as the rotor's own, added to it once the clutch has locked: half the drive's. The rotor brake and the transmission's breakaway friction hold the rotor at rest. The mass and CG are the type's, with the loads at their stations and a fuel tank that burns down.

3.9 Landing gear

Three 355×150-4 tyres [10]: their spring rates from the static deflection at the rated load, the nose wheel castoring.

3.10 FlightGear

The original aircraft's cockpit and systems (electrical, instruments, autopilot, co-pilot, crash detection, animations) stay; a bridge system maps them onto this model's properties. An in-air start holds the airframe for its first seconds while four loops trim the collective and FlightGear's trims on the vertical and angular accelerations, through the inverse of the model's control Jacobian, before letting go; the aircraft starts with its trimmed bank. For FlightGear's HDR pipeline the exterior lights are spot lights with halos.

4 Method

The model is flown offline in JSBSim's Python interface at 480 Hz. Trims solve for collective, cyclic, pedals, pitch attitude and bank (or sideslip) by Newton iteration with the airframe held and the rotor running, on the body accelerations averaged over two revolutions with a Hann window once the rotor and its wake have settled; with the rotor speed held at 353.2 rpm the engine's power is the drive's load: main and tail rotors, gearbox losses and accessories. Climbs and autorotations are trims along a sloping flight path, solved for the angle at which that power equals the power available, or nothing. Derivatives are central differences of the held airframe's loads.

A test autopilot, the UH-60 model's adapted, flies what trims cannot. It holds wheel height, ground velocity and heading in the hover; altitude, airspeed, heading by bank and sideslip in forward flight; and the rotor's speed with the collective in autorotation. Its inner loops damp the aircraft on its rates, since the Alouette has no augmentation, at crossovers chosen from the control powers. On the wheels it raises the collective steadily with the cyclic and pedals following it as a pilot's would, holds the brakes, and lets its attitude and heading loops take over once the wheels carry less than a fifth of the weight; it lowers the collective after touchdown. The same autopilot is a FlightGear autopilot of the aircraft.

5 Results

5.1 Level-flight trim and performance

Figure 1 shows the trims. At 2200 kg the model hovers out of ground effect on 364 kW and needs least, 224 kW, at 60 kt; the bank is 4.1° right in the hover, the tail rotor's push balanced. It reaches the 405 kW continuous rating at 208 km/h, against 210 km/h published as the maximum speed and V_{NE}. Past 100 kt the power rises steeply, and the last trim is at 115 kt (428 kW; 125 kt at 1750 kg): beyond it the retreating blades stall, the 63A section's lower maximum lift setting the type's speed limit about where its V_{NE} is. The longitudinal cyclic keeps a margin to its stop.

Table 4 sets the model against the published figures. The airframe's drag is reasonable (at 100 kt an equivalent flat plate of 1.4 m^2), but the performance is not: with the rated power the model would hover out of ground effect far higher than either source and climb faster. Read the other way, the power to hover at each published ceiling is the power available that ceiling implies: 342–378 kW, against the 405–420 kW ratings. At that power the model climbs at 4.5–5.9 m/s, the published 4.3–5.5 m/s, and its maximum level speed is 197 km/h, the published cruise speeds. The published figures are thus consistent with one another and with the model at about nine-tenths of the rated power: either less power reaches the Alouette's rotor than the ratings say, or the real rotor and airframe need a fifth more power than the model. Which, the public data do not tell; the model is left at the ratings and its rotor untuned, and a user who wants the published performance in FlightGear would reduce the power available, not the rotor's efficiency.

5.2 The hover and ground effect

Figure 2 shows the power to hover against altitude. At 2200 kg the hover out of ground effect can no longer be trimmed at 4877 m (it can at 3658 m): the blades reach the 63A section's stall (C_T/σ between 0.11 and 0.12) before the power runs out, so even at the ratings the model's ceiling there is its rotor's. Figure 3 shows ground effect as the test autopilot flew it: lifting off, then holding 3, 5, 10, 20, 40 and 100 ft of wheel height for half a minute each, steady to a few hundredths of a foot. At 3 ft the hover needs 0.89 of the power at 100 ft (363 kW). Each source's ceilings in ground effect imply less power available than its ceilings out of it (Aérostories 361 against 378 kW, This Day in Aviation 342 against 365 kW, at 1 m of wheel height): either their hover in ground effect is higher than a metre, or the model's ground effect, the Bo 105's and UH-60's strength and not fitted, is somewhat strong for the Alouette.

5.3 Climb and autorotation

With the 405 kW continuous rating the model climbs at 7.0 m/s at 2200 kg (50 kt) and 10.7 m/s at 1750 kg (Fig. 4), against 4.3–5.5 and 8.7 m/s published: the excess power of the previous section again. In steady autorotation the rate of descent is least, 1918 ft/min, at 60 kt.

5.4 A test flight

Figure 5 is a complete flight of 419 s, flown by the test autopilot from the wheels to the wheels. In cruise at 100 kt the engine gives 332 kW. For the practice autorotation the fuel flow lever goes to idle: the engine runs down to 16330 rpm, the freewheel lets the rotor overrun it, and the autopilot lowers the collective to hold the rotor's speed, which sags to 304 rpm at the entry, as it would with a pilot's two-second reaction, and recovers. The steady descent at 62 kt is 1982 ft/min. With the lever back to flight the engine accelerates, the freewheel takes

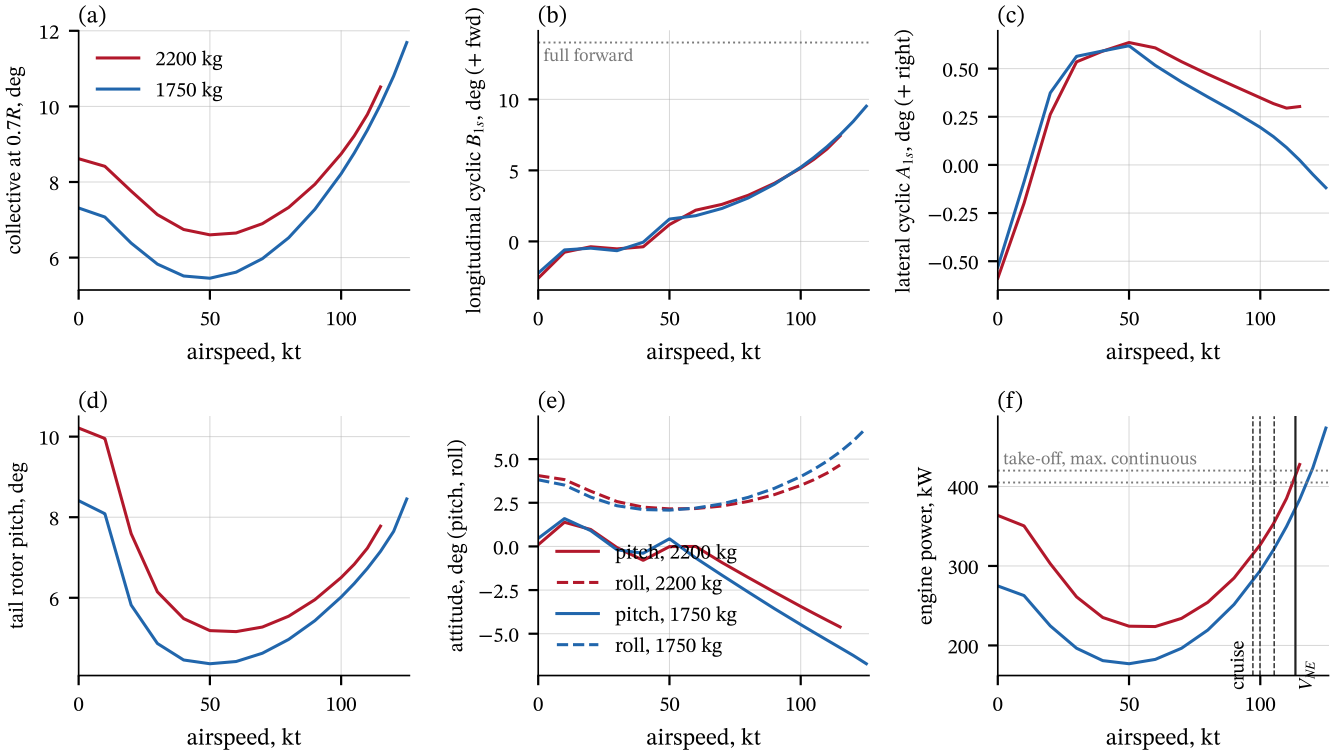


Figure 1: Level-flight trims at sea level, 2200 and 1750 kg: (a)–(d) controls, (e) attitudes, (f) the engine's power against the ratings, with the published cruise speeds (dashed) and V_{NE}.

Table 4: Performance against the published figures (AS Aérostories, TD This Day in Aviation, AO the AOPA Pilot's flight test [15]). At the published hover ceilings, the model's power to hover there: the power available that each ceiling implies.

Quantity (sea level, standard day)	Published	Model	Unit
Maximum level speed, 405 kW	210 (V_{NE})	208	km/h
Power at 185 km/h	–	326	kW
Power at V_{NE}, 210 km/h	–	414	kW
Rate of climb at 405 kW, 2200 kg	4.3 (TD), 5.5 (AS)	7.0	m/s
Rate of climb at 405 kW, 1750 kg	8.7 (TD)	10.7	m/s
Hover OGE 2200 kg: power at sea level; at the published ceilings	100 m (TD), 1100 m (AS)	364; 365 (TD), 378 (AS)	kW
Hover IGE 2200 kg: power at sea level; at the published ceilings	1650 m (TD), 2850 m (AS)	324; 342 (TD), 361 (AS)	kW
Hover OGE 1750 kg: power at sea level; at the published ceilings	1650 m (TD)	275; 286 (TD)	kW
Hover IGE 1750 kg: power at sea level; at the published ceilings	5550 m (TD)	247; 300 (TD)	kW
Rate of climb, 2200 kg, at the power the ceilings imply (342–378 kW)	4.3 (TD), 5.5 (AS)	4.5–5.9	m/s
Maximum level speed at 361 kW	180–195 (cruise)	197	km/h
Autorotation, least rate of descent, 2200 kg	about 1700 (AO)	1918 at 60 kt	ft/min

Table 5: Ground effect flown by the test autopilot, 2200 kg, sea level.

Target, ft	Wheels, ft	z/D	Power, kW	Ratio
3	2.9	0.34	322.2	0.887
5	5.0	0.40	331.3	0.912
10	10.0	0.54	344.2	0.948
20	20.0	0.82	354.7	0.976
40	40.0	1.37	360.4	0.992
100	100.0	3.03	363.2	1.000

up the drive as the spool catches the rotor, and only then does the autopilot climb, as a pilot waits for the needles to join. It lands at 13 ft/min.

5.5 Derivatives, modes and step responses

No identified derivatives of the Alouette III are public; Table 6 lists the model's. They are those of an articulated rotor with a small hinge offset under a light, narrow airframe. The roll damping L_p is $-6.24\,\mathrm{s}^{-1}$ in the hover, high for its class because the roll inertia is small; the pitch damping M_q, $-1.56\,\mathrm{s}^{-1}$, is a quarter of it, the pitch inertia being nearly five times larger. The yaw damping N_r is $-0.54\,\mathrm{s}^{-1}$ and the heave damping Z_w $-0.34\,\mathrm{s}^{-1}$. The hover's modes are the classic unaugmented helicopter's: a short-period roll subsidence and a slow, unstable pitch–speed oscillation. Figure 6 shows the responses to steps: the roll rate peaks within half a second and then falls away as the aircraft starts to slide sideways and its dihedral effect resists; the pitch rate builds more slowly and keeps building for two seconds, as the articulated rotor's lower pitch damping lets it; the yaw rate builds over three, against little yaw damping.

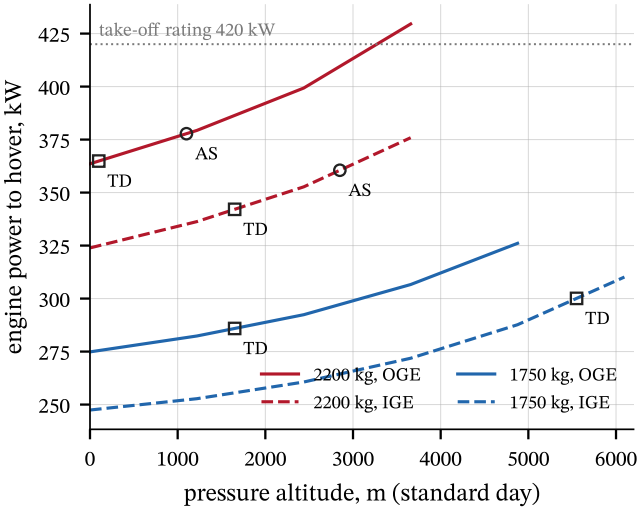


Figure 2: Power to hover against altitude, in ground effect (1 m wheel height) and out of it, with the published ceilings placed on the model's curves (AS circles, TD squares).

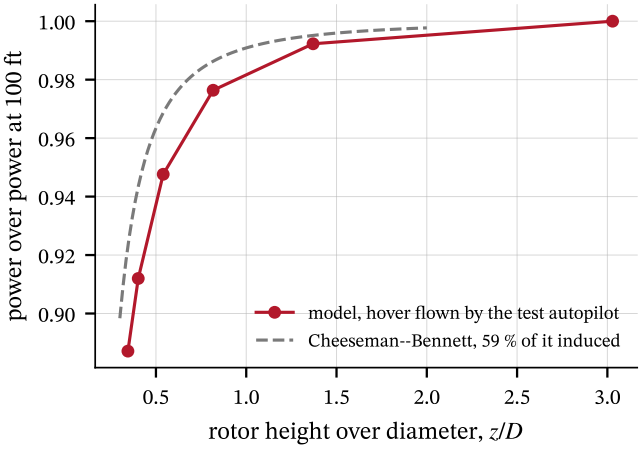


Figure 3: Ground effect: the power to hover flown by the test autopilot at each wheel height, over the power at 100 ft, against Cheeseman and Bennett's ground effect on the induced share of the power (the weight times the hover's induced velocity).

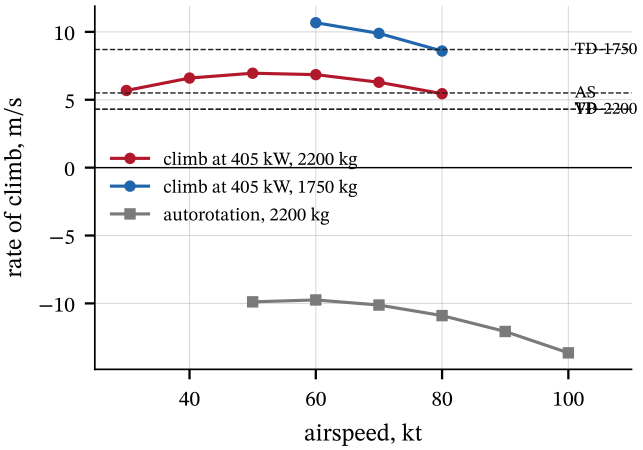


Figure 4: Rate of climb at the maximum continuous power, and steady autorotation with the rotor at 353 rpm, 1000 ft; the published climb rates dashed.

5.6 The start, the governor, a power failure

The start (Fig. 7a) runs as described [11, 12]. The engine lights off at 14.1 s, the starter drops out at 14 000 rpm and it idles at 16330 rpm (15 000–18 000 [15, 11]) and 299 °C; T4 peaks at 384 °C in the start, under the 630 °C limit. With the lever advanced the rotor starts turning at 20823 rpm of the engine (19 500–21 000 in the add-on's manual [12]), the clutch locks 83.2 s after the button and the rotor is at 95 % at 81.7 s (the lever here goes to full in 10 s, the spool governed while the clutch still slips; held at the recordings' 80 %, the rotor locks to it 44 s after it starts turning, and the spool and rotor then reach governed speed together as fast as the lever goes); T4 peaks at 404 °C as the engine carries the rotor up. Raising the collective 10 % in the hover, the power rises from 367 to 487 kW and the rotor droops 0.28 %: the single-shaft engine is the rotor's flywheel, and its governor has no droop of its own. With the power gone at 60 kt and the collective held, the rotor falls below 310 rpm in 1.0 s and to the 270 rpm limit in 2.4 s, the time a pilot has to lower the collective.

Table 6: Derivatives at 2200 kg, 1000 ft (SI, per rad; controls per % of travel), and the modes.

Derivative	0 kt	60 kt	100 kt
X_u	−0.021	−0.040	−0.060
Z_w	−0.344	−0.806	−0.899
M_u	−0.051	−0.001	+0.016
M_w	−0.025	−0.090	−0.088
M_q	−1.556	−2.228	−2.328
Y_v	−0.077	−0.071	−0.090
L_v	−0.092	−0.020	−0.015
N_v	+0.076	+0.128	+0.143
L_p	−6.245	−5.133	−4.216
L_q	+2.268	+2.627	+2.946
M_p	−0.504	−1.140	−1.006
N_p	−0.205	−0.486	−0.361
N_r	−0.543	−1.166	−1.469
Z_{col}	−24.354	−28.052	−31.885
M_{lon}	+5.427	+6.064	+6.557
L_{lat}	+13.889	+13.833	+14.079
N_{ped}	+9.269	+10.614	+13.202
N_{col}	−6.814	−4.043	−5.656
L_{ped}	+2.962	+3.351	+4.109

Speed	Modes: [damping ratio, frequency rad/s], (inverse time constant 1/s)
0 kt	[−0.06, 0.41], [0.97, 0.41], (6.04), (1.62), (0.78), (−0.40)
60 kt	[0.97, 3.07], [0.29, 1.99], [0.24, 0.06], (2.29), (0.04)
100 kt	[0.79, 3.47], [0.27, 2.69], [0.22, 0.15], (2.03), (0.04)

5.7 The vortex ring state

In descent the induced velocity follows Johnson's model (Fig. 8). The hover's induced velocity is 32.3 ft/s (1940 ft/min). In vertical descent the collective needed falls with the descent rate down to 869 ft/min and rises beyond it to 2819 ft/min: between the two, more descent needs more collective, so at a fixed collective the descent runs away, the vortex ring state's settling. At 10 kt the region is 939–2501 ft/min; from 20 kt, where the edgewise flow exceeds the induced velocity, there is none.

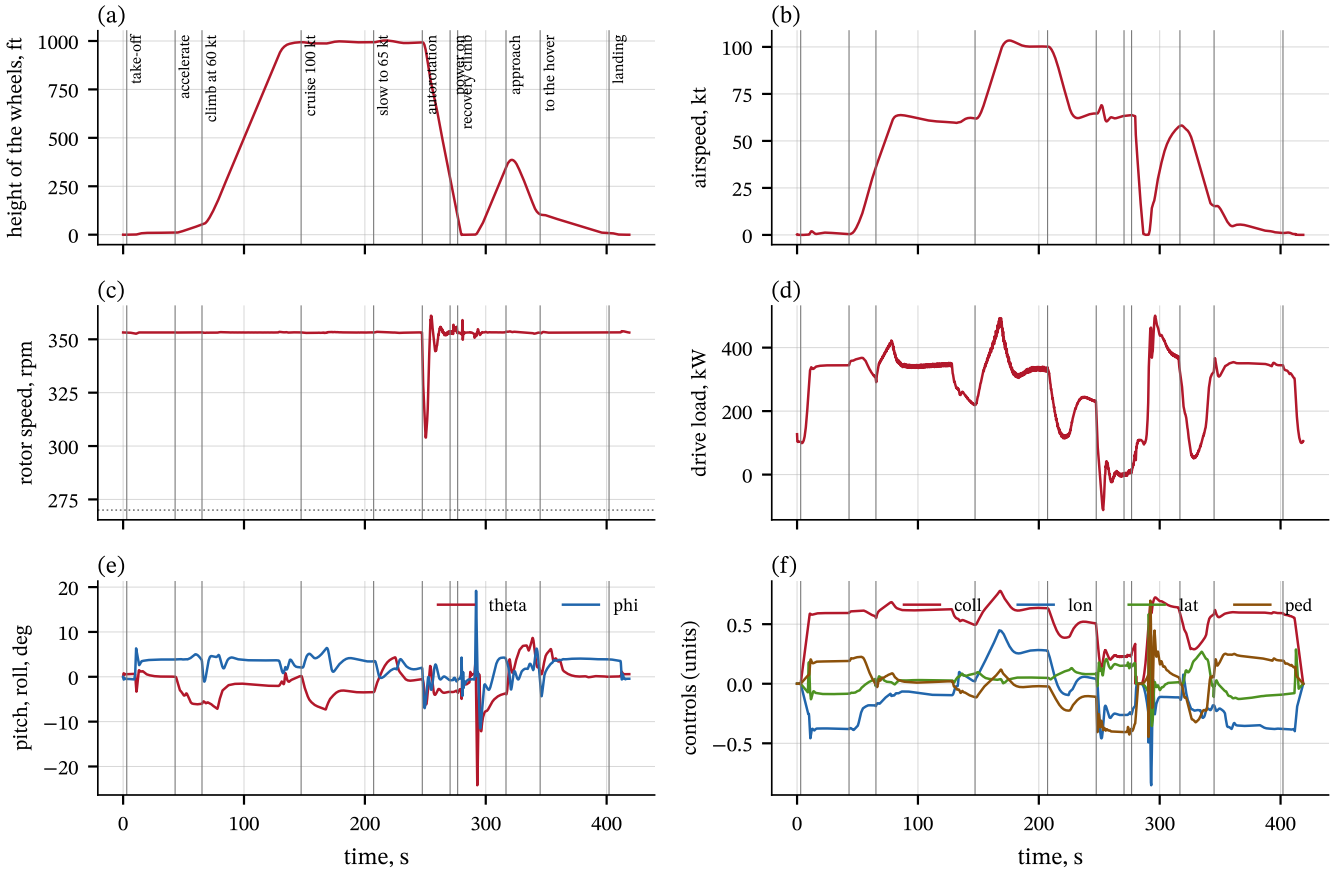


Figure 5: A test flight flown by the test autopilot, 2200 kg: take-off to a 10 ft hover, acceleration, climb to 1000 ft at 60 kt, cruise at 100 kt, a practice autorotation at 65 kt with the fuel flow lever at idle, power recovery, approach, hover, landing. The drive's load is the engine's power while the clutch is engaged; in autorotation the rotor drives the tail rotor and the gearbox itself.

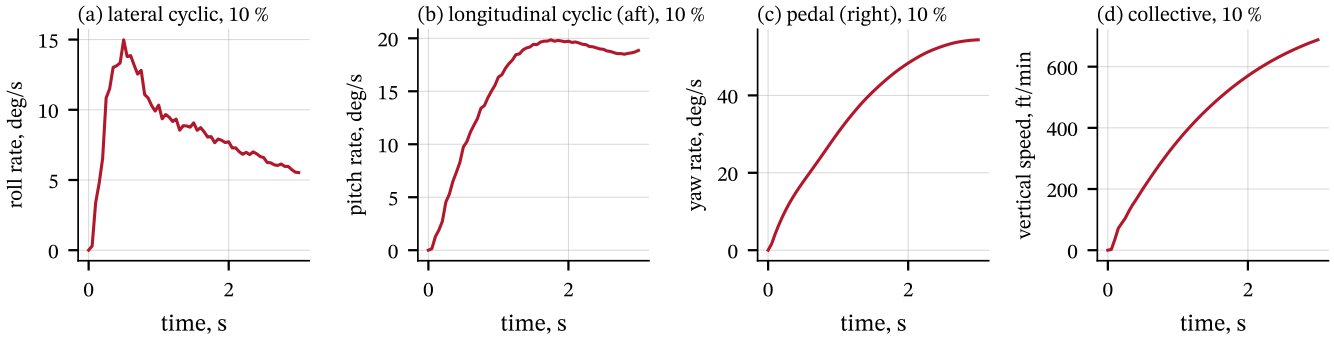


Figure 6: Responses of the free aircraft in the hover (2200 kg, out of ground effect) to 10 % of each control's travel, held.

6 Corrections found during the test programme

- *The in-air start.* FlightGear trims an aircraft that starts in the air with JSBSim's own trim, which cannot trim this rotor: the aircraft dived away. Holding the airframe while four loops trimmed the controls, at gains guessed from the control powers, let it go untrimmed: the model's control powers were three to four times the guesses, and the loops hunted. They now use the inverse of the model's control Jacobian and converge in two seconds; and since FlightGear starts the aircraft wings level, the aircraft supplies its trimmed bank.
- *The rotor at rest.* With the friction at standstill a smooth function of the rotor's speed and the speed floored at zero, the blades' swing on their hinges could ratchet a parked rotor round. A static friction now holds it while the torque from outside the rotor, from the clutch, the engine or the air, is less than the brake's and the transmission's.
- *The clutch.* As it locked, the spool's inertia entered the rotor's equation a step late: for that step the rotor's acceleration had the hub's inertia alone, a hundredth of the drive's, and the torque it put on the airframe, several hundred thousand ft lb, yawed the aircraft on its wheels; the clutch then let go and locked again. Slipping, it passed its whole capacity at any slip, and the hub, light with its blades free in lag, chattered about the spool's speed. The spool's inertia now follows the clutch's state, the shaft's torque is taken before the state changes, and the slipping clutch's torque rises over 2 rad/s of slip.
- *The electrical master* was off after an in-air start: lights and instruments had no power.
- *The test scripts* placed the aircraft on the ground at a CG height read before the model had computed its mass prop-

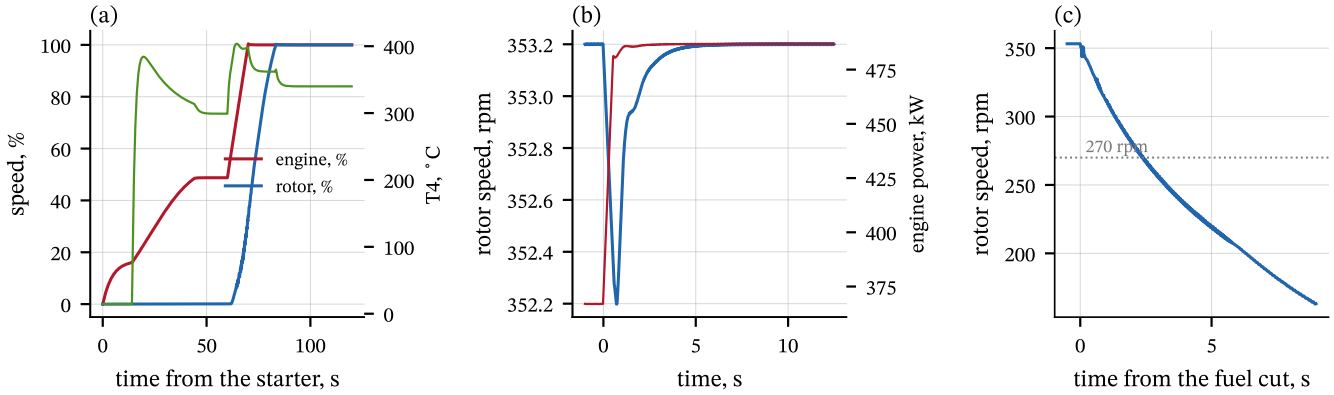


Figure 7: (a) The start: the starter pressed at 0 s, the fuel flow lever advanced from idle over 10 s at 60 s; (b) the governor, the collective raised 10 % of its travel in 0.5 s in the hover; (c) a power failure at 60 kt, the collective held.

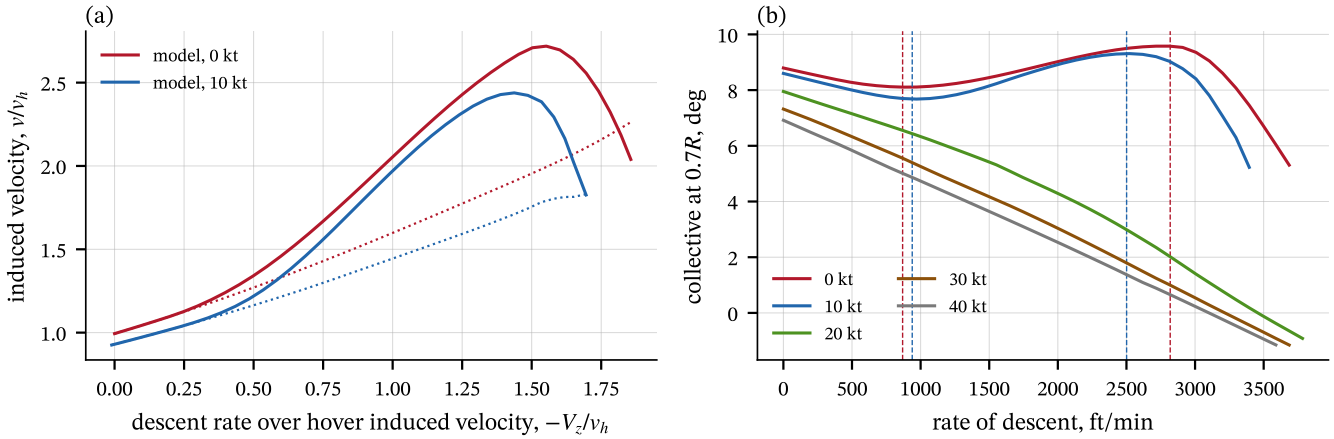


Figure 8: Steady descents at 2200 kg, 1000 ft: (a) the induced velocity against the descent rate, model (solid) and momentum theory (dotted), in vertical descent and at 10 kt; (b) the collective against the descent rate at each speed, where it turns the heave mode's stability boundary (dashed).

erties, a foot or more low; the test autopilot's wheel height had the same error.

- *The lift-off.* The UH-60's autopilot lifted the Alouette level with the hover's lateral cyclic: its tail rotor's push to the left, balanced in the hover by the thrust's lean to the right, rolled it over its left wheel within a foot. Its cyclic now comes in as the aircraft gets light, as a pilot's does, with the stick aft, the Alouette's CG being ahead of its mast.
- *Fitted parameters.* The model's comments claimed that ground effect, the miscellaneous drag and the engine's lapse were fitted to the published performance; none was, and they now say what they are.

7 Assessment

Table 7 summarises. Where the Alouette III's published data reach, the model agrees with them in its engine's behaviour and, at about nine-tenths of the rated power, in its performance; at the rated power it outperforms the aircraft, an open question whose answer lies outside the public data. Its dynamics rest on physics and on estimates (the blades' mass and inertia, the lag damper, the control ranges, the servos) for which no Alouette data are public, and nothing published checks the trims, the derivatives or the responses. The model is flown well by an autopilot with a pilot's technique, which says that it handles like a helicopter; whether it handles like this helicopter, only an Alouette pilot can say.

Table 7: Summary: ✓ reproduced, ~ partly or by construction, × not reproduced, – no data to check.

Hover, climb and speed performance at the rated power	×
The published performance with one another, at 342–378 kW	✓
Airframe drag (wind tunnel fuselage)	~
The start: idle, the clutch's engagement	✓
Rotor speed limits in autorotation	~
Ground effect	~
Vortex ring state (Johnson's model)	~
Trim controls and attitudes	–
Derivatives, modes, responses	–

8 Conclusions

- A blade-by-blade model of the Alouette III's articulated rotor, with a finite-state wake, Johnson's vortex ring state, a blade-element tail rotor, the wind-tunnel fuselage and a single-shaft Artouste with its clutch, freewheel, governor, starter and inertia, runs in plain JSBSim, generated from public data. It is the flight model of the FlightGear Alouette III.
- Its rotor turns the Alouette's way, through a mirror-image formulation of the UH-60 and Bo 105 models' equations.
- The start, idle and the clutch's engagement follow a recording, the AOPA Pilot's flight test, a maintenance description of the Lama's engine and a simulator manual after the Lama's flight manual; the Alouette III's own flight manual

was not at hand.

- The published performance is consistent with itself and with the model at 342–378 kW; at the 405–420 kW ratings the model hovers higher and climbs faster than any source gives. The discrepancy is documented, not tuned away.
- A test autopilot flies the model from lift-off through cruise and a practice autorotation to touchdown.

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